

● RESEARCH GRADE · 99%+ PURITY · USA-BASED

RESEARCH REFERENCE & SAFETY DOCUMENTATION

MOTS-c

Mitochondrial-derived peptide · Research-grade reference material

LENGTH

16 amino acids

FORMULA

$C_{101}H_{152}N_{28}O_{22}S_2$

MW

2174.6 g/mol

ENCODED BY

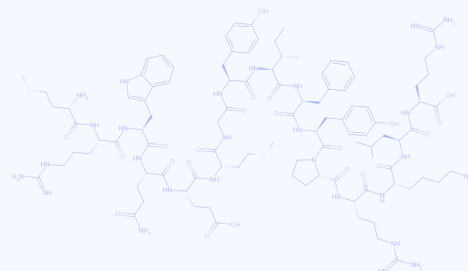
mtDNA 12S rRNA

Sold for laboratory research purposes only. Not for human or veterinary use, consumption, or therapeutic application. Must be 21+ to purchase. |
Independently verified · HPLC + mass spectrometry · Certificate of Analysis per lot.

Identity & Composition

MOTS-c (Mitochondrial ORF of the 12S rRNA Type-c) is a 16-amino-acid peptide encoded by an open reading frame within the mitochondrial 12S rRNA region — one of a class of mitochondrial-derived peptides. Highly conserved across species. Supplied lyophilized, research grade.

Product name	MOTS-c
Full name	Mitochondrial ORF of the 12S rRNA Type-c
Class	Mitochondrial-derived peptide (MDP)
Length	16 amino acids
Molecular formula	C ₁₀₁ H ₁₅₂ N ₂₈ O ₂₂ S ₂
Molecular weight	2174.6 g/mol
Encoded by	Mitochondrial 12S rRNA ORF
Appearance	White lyophilized powder
Identified use	In-vitro research & development only



Molecular structure. 2D structure computed from the primary sequence; formula and mass consistent with C₁₀₁H₁₅₂N₂₈O₂₂S₂ (2174.6 g/mol).

Met	Arg	Trp	Gln	Glu	Met	Gly	Tyr
Ile	Phe	Tyr	Pro	Arg	Lys	Leu	Arg

Primary sequence (N → C): MRWQEMGYIFYPRKLR

Note on origin. Unlike nuclear-encoded peptides, MOTS-c is encoded within the mitochondrial genome itself — part of the discovery that mitochondria produce signaling molecules, not only energy. Plasma MOTS-c levels have been reported to decline with age.

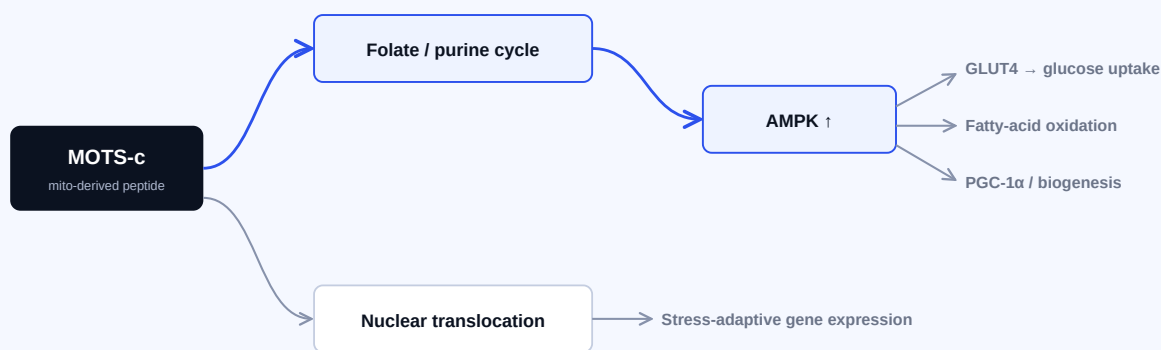
Research Background

First described by Lee et al. (2015), MOTS-c helped establish that the mitochondrial genome encodes signaling peptides — reframing mitochondria from passive energy suppliers to active participants in cellular communication. Research since has focused on metabolic regulation, exercise adaptation, and stress response.

Mechanisms studied in preclinical models

Findings below derive from cell-culture and animal systems. They are not established effects in humans.

- **AMPK activation (central mechanism).** Reported to act indirectly by inhibiting the folate cycle and de novo purine synthesis, raising AICAR and activating AMPK — distinct from direct AMPK activators.
- **Glucose handling.** Enhanced glucose flux and GLUT4-associated uptake in skeletal muscle models; reduced glucose levels in treated mice.
- **Fatty-acid oxidation & mitochondrial biogenesis.** ACC phosphorylation and PGC-1 α -associated adaptations reported in animal models.
- **Nuclear translocation.** Under metabolic stress, MOTS-c has been reported to move to the nucleus and influence nuclear gene expression.
- **Exercise interaction.** Reported synergy with exercise intervention on PGC-1 α expression and insulin resistance via AMPK signaling in mice.



Proposed pathways investigated in preclinical (in-vitro / animal) models. Schematic only; not a claim of established human effect.

What the Literature Supports — and What It Doesn't

MOTS-c has a coherent and actively replicated preclinical mechanism. The human literature is largely **observational** — associating endogenous plasma MOTS-c levels with metabolic states — rather than interventional. That distinction matters.

Evidence at a glance

PRECLINICAL MECHANISM

Well replicated

HUMAN — OBSERVATIONAL

Present (assoc.)

HUMAN — INTERVENTIONAL

Not established

REGULATORY

Not an approved drug

- Human work to date largely reports **endogenous plasma MOTS-c associations** with metabolic states (a 2024 systematic review and meta-analysis in *Diabetol Metab Syndr* examined MDP–metabolic correlations).
- Lower circulating MOTS-c has been **reported in** type 2 diabetes, gestational diabetes, and obesity cohorts — an association, not a demonstrated causal or therapeutic effect.
- Interventional data on **exogenous administration is animal-model based**; no approved human indication.
- The “exercise mimetic” description originates in **preclinical** transcriptional findings and should not be read as an established human outcome.

Appropriate characterization: a mitochondrial-derived peptide with a distinctive, well-described preclinical AMPK mechanism and consistent observational human associations, awaiting interventional human study.

Regulatory & status notes

US FDA	Not approved for any use.
Supply basis	Research reagent for in-vitro laboratory use only; not for human or veterinary administration.

Selected references

- Lee C, et al. The mitochondrial-derived peptide MOTS-c promotes metabolic homeostasis and reduces obesity and insulin resistance. *Cell Metab*. 2015.
- Zhou Q, Yin S, Lei X, et al. The correlation between mitochondrial derived peptide (MDP) and metabolic states: a systematic review and meta-analysis. *Diabetol Metab Syndr*. 2024;16:200.
- Zheng Y, Wei Z, Wang T. MOTS-c: A promising mitochondrial-derived peptide for therapeutic exploitation. *Front Endocrinol*. 2023;14:1120533.
- Yang B, et al. MOTS-c interacts synergistically with exercise intervention to regulate PGC-1 α expression... via AMPK. *BBA Mol Basis Dis*. 2021;1867:166126.
- Pham T, et al. Mitochondria-derived peptide MOTS-c restores mitochondrial respiration in type 2 diabetic heart. *Front Physiol*. 2025;16:1602271.

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	MOTS-c
Formula / MW	C101H152N28O22S2 / 2174.6
Class	Mitochondrial-derived peptide (16 aa)
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use
Supplier	
Emergency	

2 Hazard Identification

Not classified based on currently available data. Limited toxicological data; absence of classification is not a determination of absence of hazard. **Signal word:** None. **Pictograms:** None required. **Precautionary:** P261, P264, P280, P501. Handle as a potentially bioactive peptide of unknown toxicity.

3 Composition

Single-substance product, >98% research grade. Residual synthesis reagents, solvents and counter-ions may be present consistent with research-grade material. See lot COA for impurity profile.

4 First-Aid Measures

Eyes: rinse with water. **Skin:** wash with soap and water; remove contaminated clothing. **Inhalation:** fresh air. **Ingestion:** rinse mouth; do not induce vomiting unless directed. Treat symptomatically; no specific antidote known.

5 Fire-Fighting

Media: CO₂, dry chemical, foam, water spray. Combustion may produce CO, CO₂, NO_x and sulfur oxides. Use SCBA and full protective gear.

6 Accidental Release

Avoid dust; ventilate; use PPE per §8. Prevent entry to drains/waterways. Collect in a closed labeled container for disposal per §13.

7 Handling & Storage

Handle as bioactive peptide of unknown toxicity; minimize exposure. Weigh/reconstitute in fume hood. Store lyophilized solid tightly closed, protected from light/moisture/heat; -20 °C long-term, 2-8 °C short-term. Aliquot reconstituted solutions; store frozen, protected from light. Observe COA retest date.

8 Exposure Controls / PPE

No established OEL; control to ALARA. Fume hood for dry powder. Nitrile gloves (double-glove for neat powder); ANSI Z87.1 eye protection; lab coat; N100/P100 respirator where controls insufficient. Eyewash/safety shower per ANSI Z358.1.

9 Physical & Chemical Properties

State	Lyophilized powder
Appearance	White powder
Odor	Odorless
Solubility	Soluble in water
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable under normal storage. Avoid heat, moisture, sunlight, UV, open flame. Incompatible with strong oxidizers, strong acids/bases, strong reducing agents. Methionine-containing sequences are susceptible to oxidation. Decomposition: CO, CO₂, NO_x, SO_x. No hazardous polymerization.

11 Toxicological Information

Not fully characterized. No LD₅₀/LC₅₀ available. Skin/eye: treat as potentially irritating. Sensitization: cannot be ruled out. Mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: no authoritative substance-specific data; not classified based on available data; hazards cannot be excluded. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

No substance-specific ecotoxicity, persistence or bioaccumulation data identified. Avoid release to surface water, soil, drains or sewers.

13 Disposal

Dispose per applicable local/state/federal regulations; US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

14 Transport Information

Not regulated as dangerous goods under DOT (49 CFR), IATA or IMDG based on current classification. UN number: not applicable. Shipper must independently verify.

15 Regulatory Information

US TSCA: may qualify for R&D exemption (40 CFR 720.36). OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for SR&D use; verify exemption. Not a legal determination of regulatory status.

16 Other Information

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